

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 The surface integral of a vector field represents:		1
a) circulation b) divergence c) flux d) curl		
2 According to Coulomb's law, the force between two charges is:		1
a) independent of the distance b) directly proportional to the distance		
c) inversely proportional to the square of the distance d) directly proportional to the square of the distance		
3 Gauss's law is useful in calculating electric fields for _____ type of charge distributions		1
a) random b) symmetrical c) asymmetrical d) time-dependent		
4 The SI unit of magnetic flux density is:		1
a) Tesla b) Weber c) Ampere/meter d) Henry		
5 The tangential component of B across a boundary:		1
a) Is discontinuous b) Is zero		
c) Is continuous d) Depends on the current density		
6 Which Maxwell's equation relates to Faraday's law of induction?		1
a) $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ b) $\nabla \cdot \mathbf{B} = 0$ c) $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ d) $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 (\partial \mathbf{E}/\partial t)$		
7 The displacement current term in Maxwell's equations is introduced to:		1
a) Account for magnetic monopoles b) Eliminate the electric field		
c) Replace conduction current d) Satisfy the continuity equation for time-varying fields		
8 On a Smith Chart, a clockwise movement represents:		1
a) Moving toward the load b) Moving away from the load		
c) No change in the distance from the load d) Reflection coefficient approaching zero		
9 In phasor form, Maxwell's curl equation $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ becomes:		1
a) $\nabla \times \mathbf{E} = -j\omega \mathbf{B}$ b) $\nabla \times \mathbf{E} = j\omega \mathbf{B}$ c) $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ d) $\nabla \times \mathbf{E} = 0$		
10 A uniform plane wave in a lossless dielectric medium satisfies the relation:		1
a) $\nabla \cdot \mathbf{E} = 0$ b) $\nabla \cdot \mathbf{B} = 0$ c) $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ d) Both a) and b)		
11 For oblique incidence, the Brewster angle is the angle at which:		1
a) Reflection coefficient is maximum b) Transmission coefficient is zero		
c) Reflection coefficient is zero for parallel polarization d) Electric field is perpendicular to the boundary		
12 The angle θ between $\mathbf{A} = 3\mathbf{a}_x - 4\mathbf{a}_y = 3$ and $\mathbf{B} = 6\mathbf{a}_x + 8\mathbf{a}_y$ is:		1
a) 0° b) 90° c) 120° d) 180°		

Q. 2 Solve the following.		12
A) Given vectors $\mathbf{P} = 6\mathbf{a}_x + 2\mathbf{a}_y$ and $\mathbf{Q} = \mathbf{a}_x - 2\mathbf{a}_y$, calculate the following:	Apply	6
(a) $\mathbf{P} \times \mathbf{Q}$		
(b) $3\mathbf{P} + 2\mathbf{Q}$		
(c) angle between $\mathbf{P}$ and $\mathbf{Q}$		
B) State and prove Stoke's Theorem.	Remember	6
Q.3 Solve the following.		12
A) Derive the continuity equation for charge conservation. Explain the significance of the equation in the context of current flow in conductors.	Apply	6
B) State the Biot-Savart Law, calculate the magnetic field at a point due to a current I flowing through a long straight wire.	Apply	6
Q. 4 Solve Any Two of the following.		12
A) State Faraday's Law of Induction. Derive the mathematical expression for the induced emf in a coil of wire when subjected to a changing magnetic field.	Understand	6
B) State and explain the gradient of a scalar field and find the gradient of the scalar field $U = x^2 y + x y z$	Understand	6
C) Derive the boundary conditions for the electric and magnetic fields at the interface between two different media.	Analyze	6
Q.5 Solve Any Two of the following.		12
A) Derive the general wave equations for voltage and current on a transmission line. Discuss the role of the characteristic impedance in these equations.	Analyze	6
B) Define (i) VSWR and (ii) reflection coefficient. A transmission line with a characteristic impedance of $Z_0 = 50\Omega$ is terminated with a load impedance of $Z_L = 75\Omega$. 1. Calculate the reflection coefficient at the load. 2. Determine the VSWR for this system.	Apply	6
C) Explain the concept of the Smith Chart and its application in the analysis of transmission lines.	Understand	6
Q. 6 Solve Any Two of the following.		12
A) State Maxwell's equations in phasor form for time-harmonic fields. Discuss the significance of each equation in describing the behavior of electric and magnetic fields in space and time.	Understand	6
B) Explain the concept of polarization in electromagnetic waves. Differentiate between linear, circular, and elliptical polarization.	Understand	6
C) A uniform plane wave of 75 Hz propagating through Copper conductor having $\sigma = 7.2 \times 10^7 \text{ S/m}$, $\epsilon_r = 1$, $\mu_r = 1$. Find (i) attenuation constant, (ii) phase constant, (iii) propagation constant, (iv) intrinsic impedance, (v) wavelength and (vi) phase velocity.	Evaluate	6

*** End ***